

Justin Manoj

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EDUCATION

University of Massachusetts Amherst

Amherst, MA

B.S. in Computer Science and Mathematics | GPA: 3.78/4.00

Expected Graduation: May 2027

- **Relevant Coursework:** Algorithms & Data Structures, Operating Systems, Database Systems, Machine Learning, Computer Networks & Security, Web Programming, Scalable Web Systems, Discrete Mathematics
- **Awards/Honors:** Dean's List, Chancellor's Award, NYS Academic Excellence Award, AP Scholar with Distinction, National Merit Scholar, Commonwealth Honors College

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Go, JavaScript, TypeScript, SQL, HTML/CSS

Technologies: Spring Boot, FastAPI, React, Node.js, gRPC, PostgreSQL, Redis, Kafka, Docker, Kubernetes

Tools/Frameworks: AWS (EC2, EKS, S3), Linux/Unix, Git, CI/CD

EXPERIENCE

ChitChat Workplace | Co-Founder / Software Engineer

May 2025 – Present

- Built real-time messaging platform in **Go** and **WebSockets**, scaling to **100,000+** concurrent users and onboarding initial enterprise customers within **3 months** of launch.
- Designed event pipeline with **Apache Kafka** and **Redis Pub/Sub**, processing **2M+** daily events reliably across messages, reactions, and typing indicators.
- Containerized services with **Docker** and deployed on **Kubernetes (EKS)**, enabling autoscaling and zero-downtime rolling updates across all platform services.
- Reduced repetitive user queries by **40%** by integrating **LangChain** with **pgvector**, enabling semantic search and AI-assisted drafting across the platform.

GBCS Group | Software Engineering Intern

March 2025 – September 2025

- Migrated **8** REST endpoints to a **GraphQL** API with **DataLoader** batching and **gRPC**-compatible internal services, cutting database queries by **60%** and reducing response payload sizes by **40%**.
- Reduced load times by **65%** using **PostgreSQL** composite indexes & **Redis** caching on high-traffic pages.
- Built **GitHub Actions** CI/CD pipeline with **Jest** test suites and **Sentry** monitoring, reducing production bugs reported per sprint by **35%**.

University of Massachusetts Amherst | Undergraduate Research Assistant

November 2024 – April 2025

- Stress-tested **7** multi-agent LLM frameworks uncovering **14** failure patterns across **10,000+** simulation runs.
- Sped up log processing **3×** with **Ray** & **Pandas**, analyzing **1,600+** execution traces to benchmark agent reliability.
- Cut review time by **70%** with an automated pipeline that classifies agent failures and surfaces critical patterns.

BUILD UMass | Software Engineer

February 2024 – May 2024

- Indexed **1,500+** campus events from **300+** student organizations using **Elasticsearch**, reducing search latency by **89%** and supporting real-time filtering by date, category, and location.
- Engineered real-time event updates with **WebSockets** and **Redis Pub/Sub**, delivering live notifications to concurrent users with **sub-50ms** delivery time.
- Worked with a team of **4** to ship to **6,000+** active UMass students, processing **500+** daily searches in the semester.

PROJECTS

UMass Dining Engine | Java, Spring Boot, React, PostgreSQL, Redis, Python

Jan. 2026

- Built AI meal planning platform for **30,000+** UMass students aggregating live menus from **4** dining halls with real-time dietary filters; pending adoption into the official UMass Dining App.
- Designed REST API with **Spring Boot** and **Redis** caching, backed by **Python** scraping pipeline for live accuracy.

Poker IQ Trainer | TypeScript, React, WebAssembly, Node.js

Apr. 2025

- Built a Texas Hold'em trainer with Beginner and Expert AI opponents using **Monte Carlo** simulations, compiled to **WebAssembly** with precomputed hand equity tables for real-time pot odds and EV training.

Medical Insight Bot | Python, FastAPI, React, PostgreSQL, pgvector

Jan. 2025

- Built a **RAG**-based symptom analysis chatbot using **pgvector** and **FastAPI**, returning ranked diagnoses with confidence scores; evaluated **GPT-4o** and **Gemini** across clinical test cases.